

Embedded Systems -- Guided Notes

Unit tp-2.13 | CompTIA Network+ N10-009 Obj. FC0-U71 2.13 | N10-008 FC0-U71 2.13

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

An embedded system is a _____ computing system built into a device for a specific _____ function. Unlike a general-purpose computer, an embedded system is not designed to run _____ different applications. Examples include _____, _____, and _____ (name three devices).

Question 2

A microcontroller is the primary chip inside many embedded systems. It integrates a _____ (processor), _____ (program storage), _____ (working memory), and I/O interfaces all into a single _____. This makes it compact, cheap, and power-efficient for dedicated tasks.

Question 3

An RTOS (Real-Time Operating System) is designed to process inputs and produce outputs within a _____ deadline. Hard real-time systems must meet every deadline or the system _____. Examples include engine control units in cars and medical _____ where a missed deadline could cause harm.

Question 4

Firmware is software stored in _____ (non-volatile memory, often flash or ROM) inside a hardware device. It controls the basic operation of the device at a _____ level. Unlike standard software, firmware _____ (can/cannot) usually be updated without special tools or vendor authorization, and updates carry _____ risk if interrupted.

Question 5

Embedded systems are common in _____ (vehicles with computerized fuel and emissions control), _____ (insulin pumps, defibrillators), _____ (assembly robots, PLCs), and _____ (routers, switches). In critical systems, embedded software is subject to rigorous _____ testing before deployment.

Question 6

The three main constraints of embedded systems that limit them compared to general-purpose computers are limited _____ (CPU speed), limited _____ (RAM and flash), and limited _____ (battery or power supply). These constraints require developers to write highly _____ code with minimal overhead.

Question 7

An SoC (System on a Chip) integrates all major components of a computer -- _____, _____, and _____ controllers -- into a single silicon die. SoCs are common in _____ phones, _____ devices, and embedded systems because they reduce size, cost, and power consumption.

Question 8 -- Real World Application

REAL WORLD: A car manufacturer recalls 50,000 vehicles because of a critical software bug in the embedded engine control unit (ECU) that can cause unintended acceleration. Unlike a PC software bug, the fix cannot simply be emailed to users. Explain what challenges the manufacturer faces in updating the embedded firmware and describe two methods they could use to deliver the fix.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-2-13-embedded-systems/tp-2-13-embedded-systems-answer-key.pdf
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